

Effect of structural relaxation on magnetism of twisted CrI_3 bilayer

Aditya Putatunda¹, Sergey Artyukhin¹

aditya.putatunda@iit.it

¹ Italian Institute of Technology, Genova, Italy,



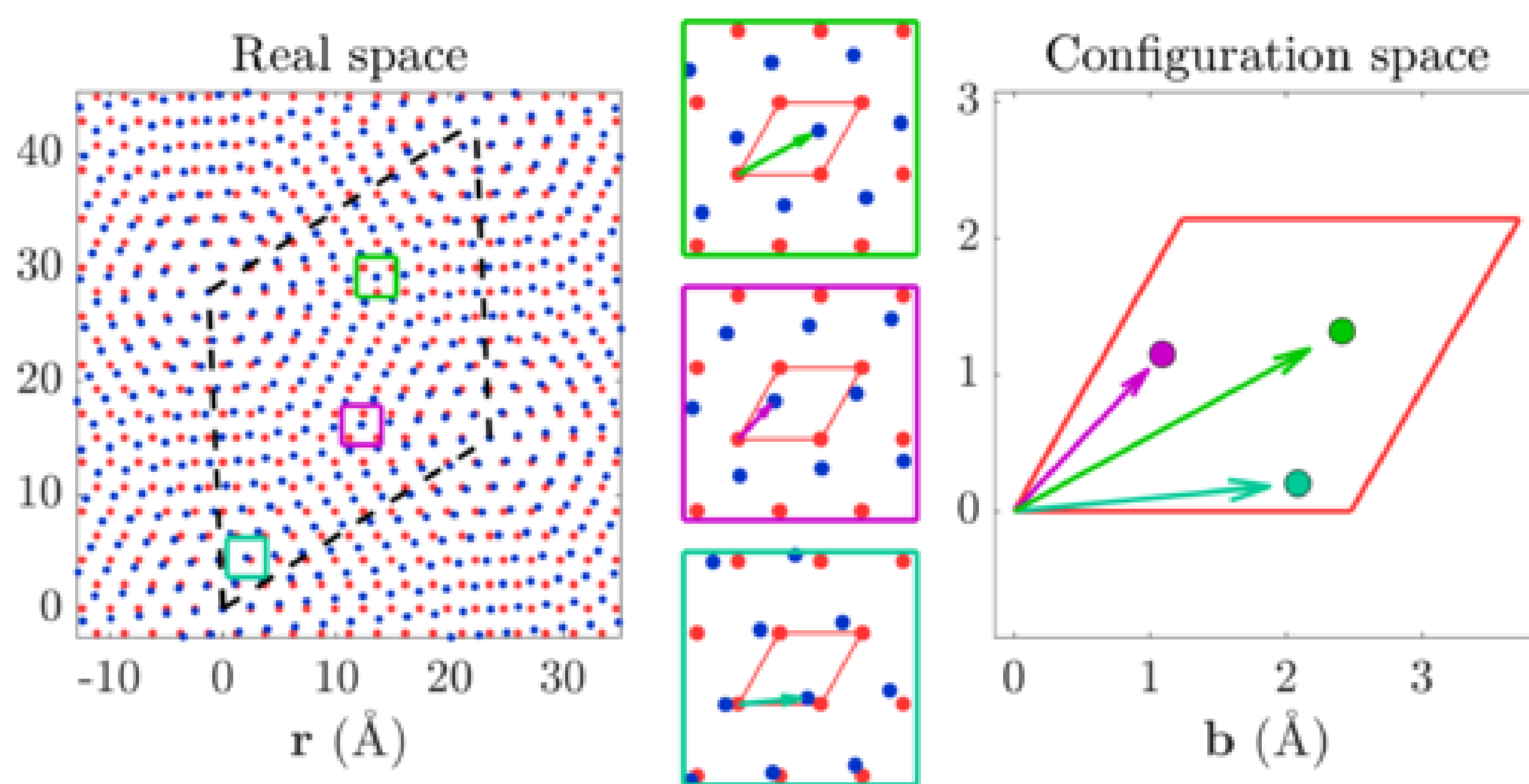
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ABSTRACT & MOTIVATION

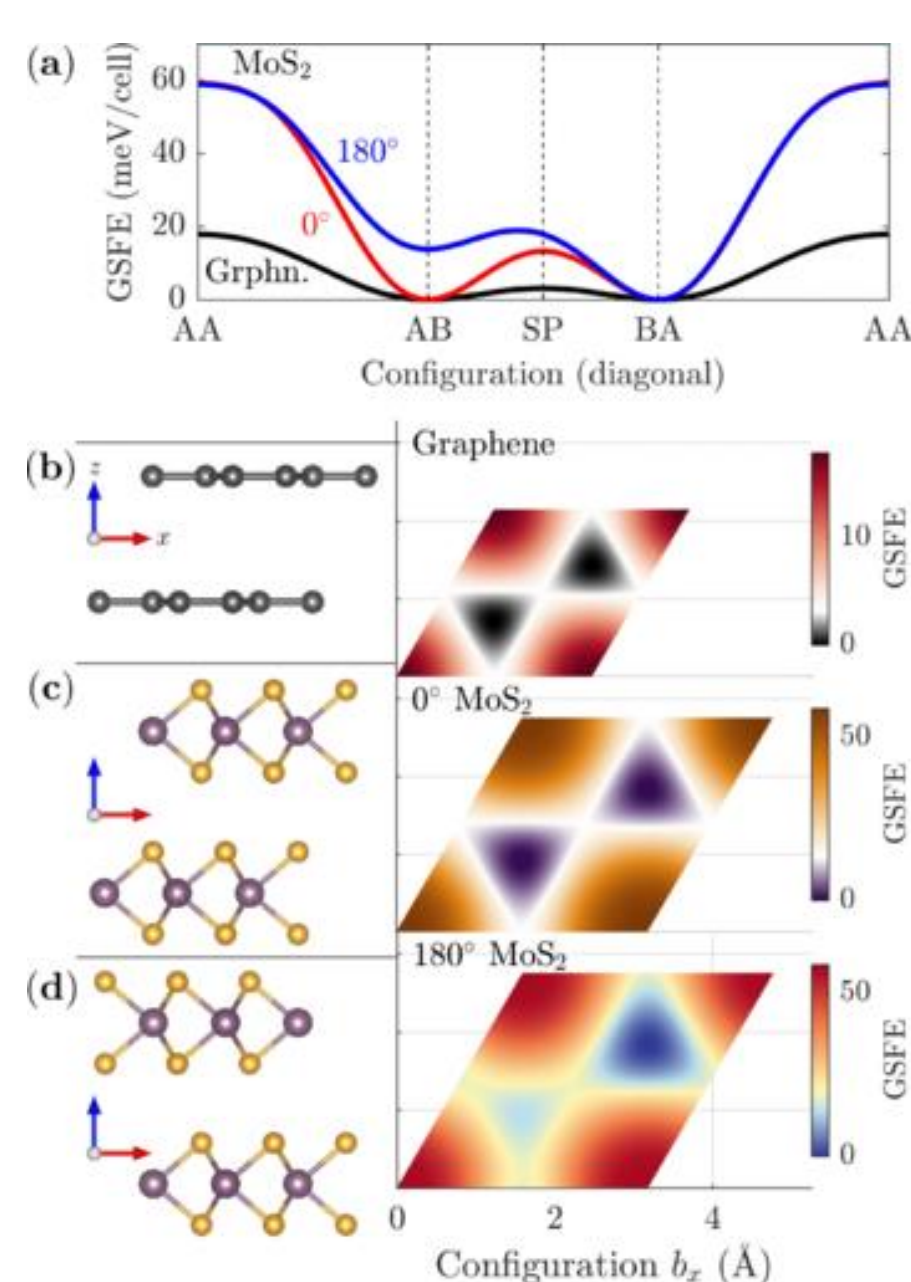
Twisted bilayer systems attract major attention for their ability to offer strongly correlated states for certain values of the twist angle. Different types of layer stacking have very different energies, which leads to the reconstruction of the Moiré patterns into regions with optimal stacking separated by domain walls with higher-energy one [Kaxiras PRR 2018]. Thus, the ionic relaxation has a profound consequences on the lattice and the electronic structure as a result. Here we explore the effect of ionic relaxation on magnetism in twisted bilayer of CrI_3 .

IONIC RELAXATION

Seminal paper [1]: Map ionic displacements of a Moiré cell in real space to configuration space. Stacking environments are described by vectors that lie within the unit cell of the red layer, allowing one to translate every atom in real space to a specific point b in configuration space.

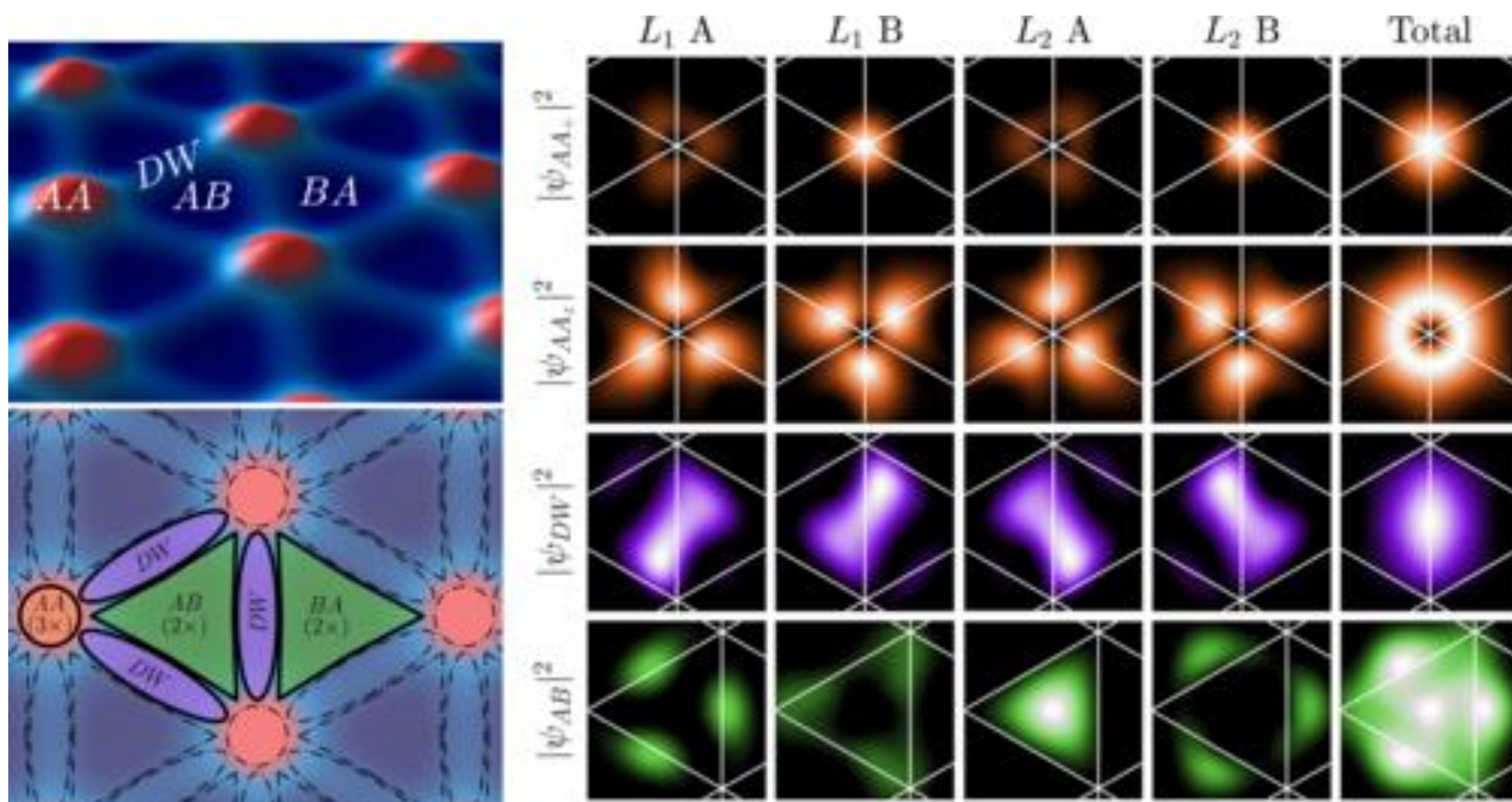


Energetics of stacking patterns



$$\begin{aligned} \mathcal{E}(\nabla U_i) &= \frac{1}{2}(\nabla U_i + \nabla U_i^T) \\ E_{\text{intra}} &= \lim_{R \rightarrow \infty} \frac{1}{|B_R|} \int_{B_R} \frac{1}{2} \mathcal{E}(\nabla U_i) C_i \mathcal{E}(\nabla U_i) dr \\ &= \lim_{R \rightarrow \infty} \frac{1}{|B_R|} \int_{B_R} \frac{1}{2} [K_i (\partial_x U_{ix} + \partial_y U_{iy})^2 \\ &\quad + G_i [(\partial_x U_{ix} - \partial_y U_{iy})^2 + (\partial_x U_{iy} + \partial_y U_{ix})^2]] dr \end{aligned}$$

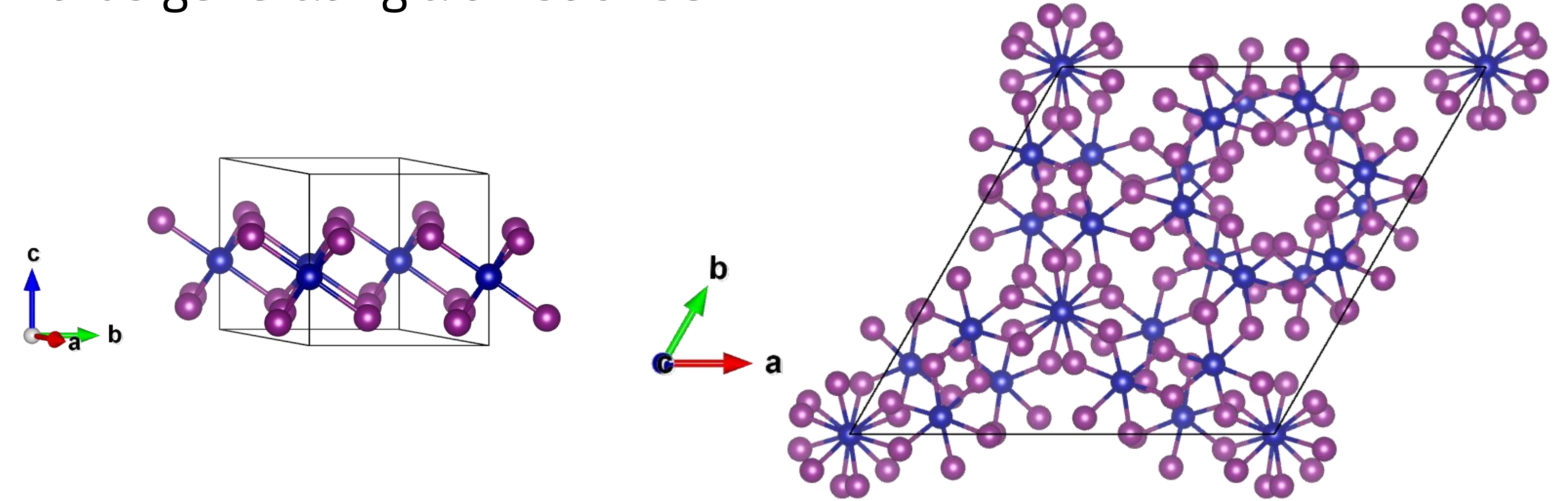
Domains with lowest energy stacking, separated by domain walls



Domains and walls host different electronic states, which may affect magnetism [2]

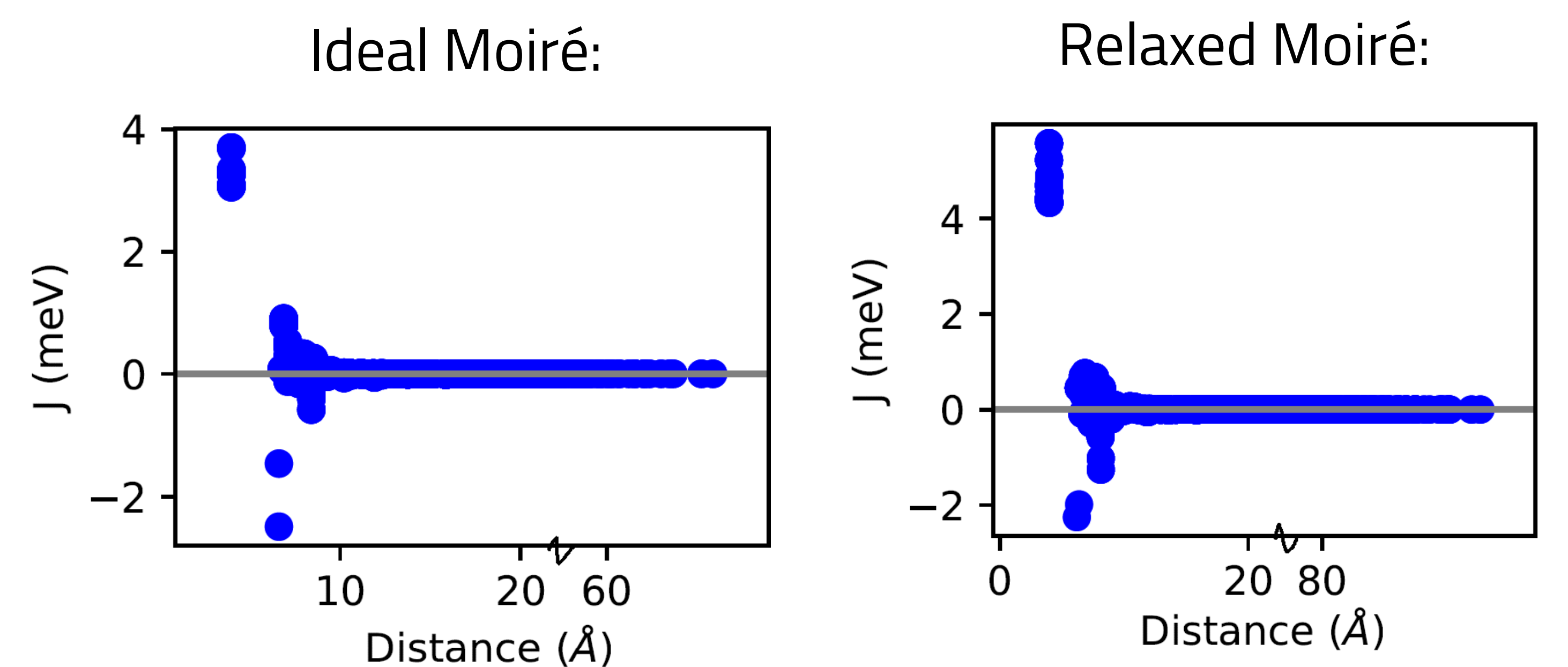
MAKING THE TWIST

Create a twisted cell: rotate two layers by angles $\theta/2$ in opposite directions to superpose ions (m,n) of one layer with $(m,-n)$ of the other. Moiré is created by superposing $(2,1)$ ions on L1 into $(2,-1)$ on L2 constructed out of CrI_3 rhombohedral cell, thus generating a twist of 38.21°



Ionic relaxation, Siesta with PBE+SOC, electronic state with $U = 3\text{eV}$; forces below $0.01\text{eV/\AA}$

RESULTS



Nearest neighbor FM (DM , J_{ani}) exchanges are enhanced (upto 20%)

RELAXATION-INDUCED TOPOLOGICAL MAGNETISM

10x10x1 SC, 2800 Cr atoms



$T = 0.0001\text{K}$ ($\Delta E_{\text{FM}} = 0.0082\text{ mRy/SC}$)

$T = 0.1\text{K}$ ($\Delta E_{\text{FM}} = 0.0164\text{ mRy/SC}$)

CONCLUSIONS & OUTLOOK

- Relaxation has a profound effect on the magnetism. Exchange constants are enhanced
- These enhanced exchanges support metastable skyrmionic spin texture against canted FM ground state
- Outlook: Include the effect of van der Waals' dispersion forces
- Smaller twisting angles
- Effects of anisotropic exchanges
- Study the topological properties of the states, phase diagram

REFERENCES

- [1] S. Carr, D. Massatt, S. B. Torrisi, P. Cazeaux, M. Luskin, E. Kaxiras, Phys. Rev. B **98**, 224102 (2018)
- [2] S. Carr, S. Fang, Z. Zhu, and E. Kaxiras Phys. Rev. Research **1**, 013001 (2019)